Yueqi Song's Statement of Purpose

My research interest is in **Natural Language Processing (NLP)**, with a focus on **creating inclusive and robust AI agents capable of assisting diverse populations worldwide by understanding and interacting with the world multimodally**. I aim to create agents that excel in three critical areas: 1) ***inclusivity*** to bridge the gaps created by the current English- and Western-centric focus of NLP, enabling equitable access for linguistically and culturally diverse communities; 2) ***multimodal support*** to perceive and generate content across various modalities such as text and images for versatile interactions; and 3) ***robust task-solving capabilities*** to provide accurate and meaningful assistance in real-world contexts. By advancing these areas, my work seeks to create AI agents that are not only powerful but also universally accessible and impactful, ensuring they serve as transformative tools for the global community.

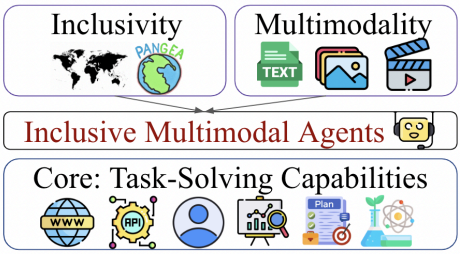


Figure 1: Overview of My Research

Inclusivity. How equal and inclusive is the current state of multilingual NLP? It is essential to ensure inclusive support in NLP, such that technological advancements benefit people from all linguistic, cultural, and socio-economic backgrounds. Promoting equity and inclusion in NLP technology is closely tied to first understanding the current state of multilingual NLP, not only models' multilingual performances but also the amount of NLP resources available across languages. To address this, I took the lead in creating a multilingual benchmark named ***GlobalBench*** to track global progress in NLP [1]. GlobalBench revealed significant disparities among languages, with under-served languages like Arabic and Portuguese experiencing lower performance and limited resources, despite their large speaker populations. This work highlighted the need for inclusive NLP systems and multilingual resources that address these inequities.

Multimodal Support. Building upon the text-only GlobalBench, I also am researching inclusive multimodal models and resources [2-4]. Inclusive support across modalities is critical for enabling models to understand complex environments and respond effectively to diverse users. In my work on ***multilingual visual reasoning*** [2], we found models struggle with effective visual reasoning across lower-resource languages. In another study on ***image transcreation*** [3], which won the *Best Paper Award at EMNLP 2024*, we found that generative models struggle to translate images culturally and often exhibit stereotypes. These studies highlight the need for models with multimodal support that prioritize multilingual inclusivity and cultural sensitivity. To further push the boundaries of universally accessible systems with multimodal support, I co-led a project ***Pangea*** [4], a fully open multilingual multimodal language model aimed at bridging multilingual and multicultural gaps in visual understanding tasks. In this project, we developed a six million multilingual multicultural instruction tuning dataset PangeaIns, and curated a comprehensive multilingual multimodal benchmark PangeaBench, to address the lack of multilingual resources. Pangea-7B supports 39 languages including many low-resource languages like Swahili and Tamil, and achieves 10% higher average performance on 14 multilingual benchmarks than existing open models, highlighting its ability to provide more inclusive support to a broader audience. With good multilingual visual reasoning and cultural awareness, Pangea is a foundational effort in addressing inclusivity and multimodality, reinforcing my commitment to creating universally accessible AI multimodal agents.

Robust Task-Solving Capabilities. I believe robust task-solving capabilities are essential for AI agents to effectively serve people. Even agents focused on interaction require strong problem-solving skills to provide accurate and meaningful assistance. Building on my foundation in multilingual and multimodal research, I expanded my focus to developing generalist AI agents with advanced task-solving abilities. I started by

contributing to the development of ***OpenHands*** [5], an open-source platform for building AI agents capable of interacting with the world through writing code, executing commands, and browsing the web. This work deepened my understanding of designing generalist AI systems capable of tackling complex tasks across diverse domains, and I began to explore the unique challenges and opportunities presented by task-solving in specific environments, such as the web. Most web agents, like those in OpenHands, simulate human interactions with graphical user interfaces (GUIs) using keyboard and mouse operations. However, is simulating human behavior the optimal solution for web agents, given the complex nature of browsing via GUIs? In my work on ***API-based web agents*** [6], I developed an agent that directly interacts with web services via APIs, designed specifically for machine-to-machine communication. We extended this to a hybrid agent that combines API calls and web browsing. Our evaluations showed that the API-based agent outperformed the GUI-based agent, while the hybrid outperformed both. These results highlight the importance of tailoring agents to their unique strengths, optimizing performance and efficiency rather than strictly mimicking human behavior.

Future Research Goals. All these prior experiences have inspired me to pursue a Ph.D. in NLP with the vision of bridging cutting-edge research and real-world impact through ***robust and inclusive multimodal AI agents that serve people worldwide***. While most generalist agents are limited to English-centric, text-only settings, I aim to push the boundaries of AI systems by ***advancing multilingual and multicultural inclusivity***, ***developing robust multimodal NLP tools***, and ***building adaptable generalist AI agents with strong task-solving capabilities***. My goal is to bridge the gap between inclusivity, multimodal tools, and unified agents, creating AI agents capable of addressing diverse, real-world challenges. I am also excited to explore other interesting directions, such as building agents with commensense, designing interactive NLP tools within social contexts, and creating user-centric systems with societal impact.

Beyond research, I value collaboration, effective communication, and teaching. As a lead Teaching Assistant for CMU's Computer Systems course, I refined my skills in presenting, mentoring, and developing learning materials. I have also presented my research at conferences and served as a reviewer for conference papers. I look forward to continuing these activities in graduate school, fostering interactive learning from and sharing insights with other amazing people to advance the field.

CMU is my preferred choice for graduate studies, specifically because there are professors who have excellent projects on interesting topics and whom I share similar interests with. I am particularly excited about Professor Graham Neubig's research on multilingual language processing and generalist code agents; my interests highly align with Professor Daniel Fried's interests on multi-agent systems and grounded interactions under task-oriented settings; I am inspired by Professor Sherry Tongshuang Wu's work on evaluating AI on real-world scenarios; and I share similar interests with Professor Chenyan Xiong on multimodal GenAI and training through data-centric approaches. Moreover, I aim to collaborate with diverse communities, leveraging their insights to develop NLP solutions that can adapt to different languages, cultures, and communication styles. CMU, with its exceptional faculty and student body, can light my way to explore relentlessly with curiosity, think unconventionally by looking outward, and work towards solving the world's major challenges in the future. Therefore, I am confident that CMU is a great place that can prepare me to achieve my career and personal goals.

References

[1] ***Y. Song***, S. Khanuja, P. Liu, F. Faisal, A. Ostapenko, G. Winata, A. Aji, S. Cahyawijaya, Y. Tsvetkov, A. Anastasopoulos, and G. Neubig, "GlobalBench: A Benchmark for Global Progress in Natural Language Processing," *Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2023. [PDF].
[2] ***Y. Song***, S. Khanuja, and G. Neubig, "What Is Missing in Multilingual Visual Reasoning and How to Fix It," *Under Review*, 2024. [PDF]
[3] S. Khanuja, S. Ramamoorthy, ***Y. Song***, and G. Neubig, "An image speaks a thousand words, but can everyone listen? On image transcreation for cultural relevance," *Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2024. [PDF].
[4] X. Yue*, ***Y. Song****, A. Asai, S. Kim, J. D. D. Nyandwi, S. Khanuja, A. Kantharuban, L. Sutawika, S. Ramamoorthy, and G. Neubig, "Pangea: A Fully Open Multilingual Multimodal LLM for 39 Languages," *Under Review*, 2024. [PDF].
[5] X. Wang, B. Li, Y. Song, F. F. Xu, X. Tang, M. Zhuge, J. Pan, ***Y. Song***, B. Li, J. Singh, H. H. Tran, F. Li, R. Ma, M. Zheng, B. Qian, Y. Shao, N. Muennighoff, Y. Zhang, B. Hui, J. Lin, R. Brennan, H. Peng, H. Ji, and G. Neubig, "OpenHands: An Open Platform for AI Software Developers as Generalist Agents," *Under Review*, 2024. [PDF].
[6] ***Y. Song***, F. F. Xu, S. Zhou, and G. Neubig, "Beyond Browsing: API-Based Web Agents," *Under Review*, 2024. [PDF].